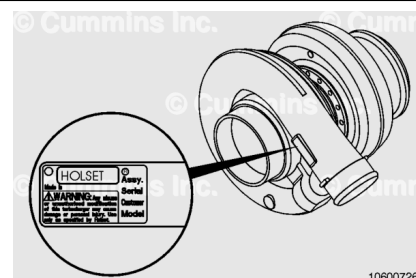


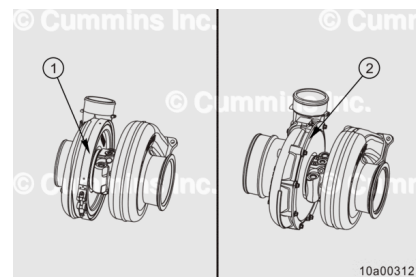
Trainings ▾

General Information

The turbochargers used on the QSK45 and QSK60 engines may differ in the service methods based on their applications. Locate the turbocharger dataplate to verify the procedures contained within this procedure apply and that the correct specifications are used.



The HE850 model name is unable to be retroactively applied to early design releases due to regulatory compliance constraints. These HE850 turbochargers will still have HE800 as the model name listed on the turbocharger dataplate. For differentiation from the standard HE800 turbocharger, the HE850 model name will be referenced throughout this procedure.



The procedure and specifications will be different based on the type of turbocharger. Identify the type of turbocharger equipped:

- The HE800 turbocharger (1) can be identified by the V-band joint between the compressor and bearing housing.
- The HE850 turbocharger (2) can be identified by the bolted joint between the compressor and bearing housing.

One turbocharger is mounted directly on the exhaust manifold.

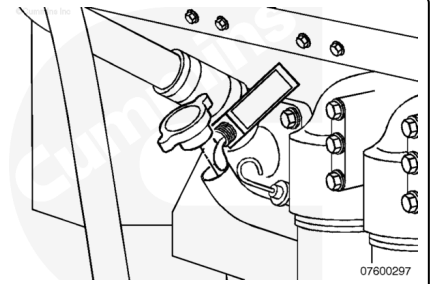
Both the HE800 turbocharger and HE850 turbocharger are cooled using coolant from the high-temperature circuit.

Each HE800 turbocharger weighs approximately 57 kg [126 lb]. Each HE850 turbocharger weighs approximately 65 kg [143 lb]. Both will require proper lifting equipment.

Fluorescent Dye Tracer

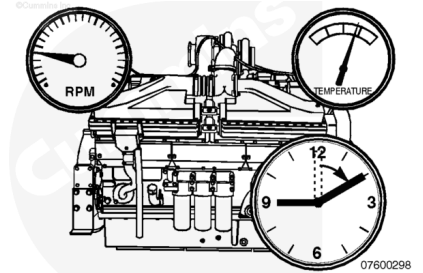
This procedure can be easily done on the QSK45 engine, but it can involve much more time and/or cost with QSK60 engines.

Add 1 unit of fluorescent tracer, Part Number 3376891, or equivalent, to each 38 liters [10 gal] of engine lubricating oil.



Operate the engine at low idle for 10 minutes.

Shut the engine OFF.



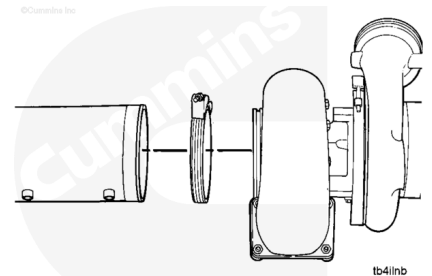
Allow the turbocharger to cool. Remove the exhaust pipe from the turbine housing.

Remove the aftercooler heat shields if present. Refer to Procedure 010-129 in Section 10.

(</qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html>)

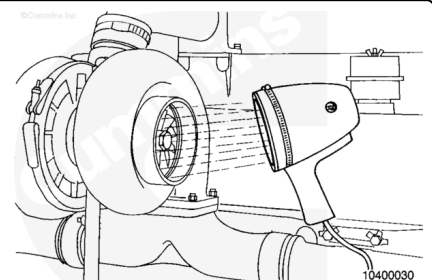
Remove the turbocharger heat shields if present. Refer to Procedure 010-076 in Section 10.

(</qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html>)



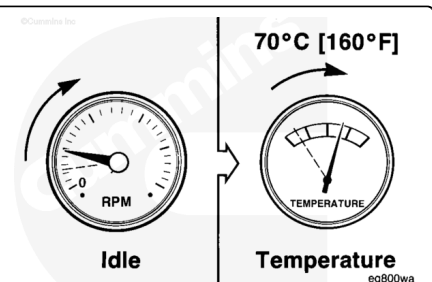
Use a high-intensity black light, Part Number 3824524 or 3824754, to inspect the turbine housing outlet for oil.

A dark blue glow indicates fuel slobber. A yellow glow indicates oil carryover.

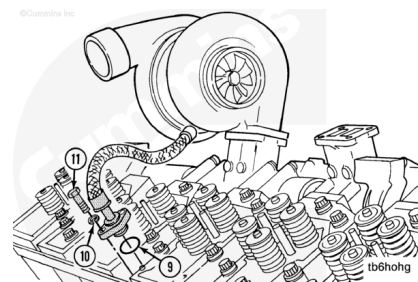


The coolant temperature needs to be maintained at a minimum of 71°C [160°F].

If fuel is found in the turbine housing, the problem is excess, unburned fuel. To correct the problem, decrease the idle period, or increase the idle rpm.



If oil is found in the turbine housing, disconnect the oil drain tube from the cylinder block.



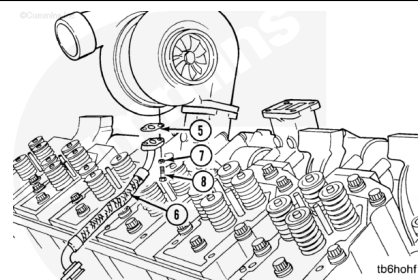
Disconnect the oil drain tube (6) from the turbocharger.

Remove any support clamps and the hose.

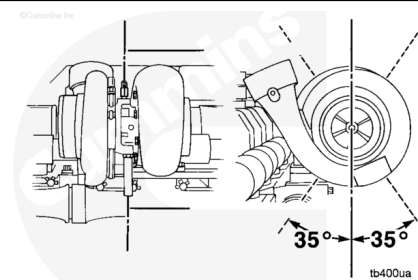
Remove and discard the o-ring gasket (5).

Clear any restrictions found in the hose.

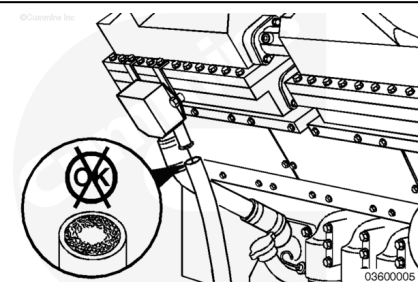
Discard the hose if it is cracked, frayed, or will **not** bend without cracking.



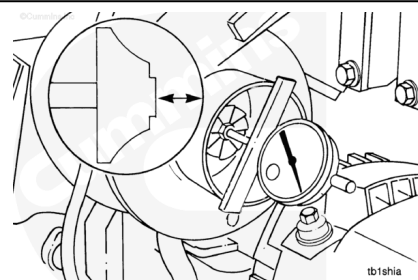
Check the angle of the drain tube. The angle of the tube **must** be within 35 degrees of vertical. Adjust the turbocharger, if necessary.



If the drain tube is free of restrictions and at the correct angle, check the crankcase breather and tube to verify they are **not** plugged or restricted.



Check the turbocharger axial clearance and radial clearance. Refer to the Inspect for Reuse section in this procedure.



Install the exhaust pipe into the turbine housing.

Install the turbocharger heat shields if present. Refer to Procedure 010-076 in Section 10.

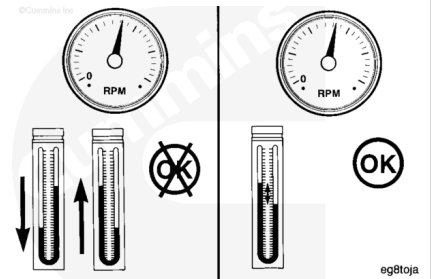
(/qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html)

Install the aftercooler heat shields if present. Refer to Procedure 010-129 in Section 10.

(/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html)

If these checks do **not** reveal the problem, measure the crankcase (blowby). Refer to Procedure 014-005 in Section 14.

(/qs3/pubsys2/xml/en/procedures/56/56-014-005-tr.html)

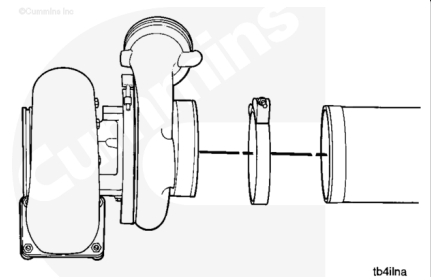


Inspect for Reuse

Note : After a turbocharger malfunction, check all components downstream for debris.

Note : The configuration illustrated is similar to the QSK45 engine. The same procedure is to be used for the QSK60 and the QSK60 two-stage engines.

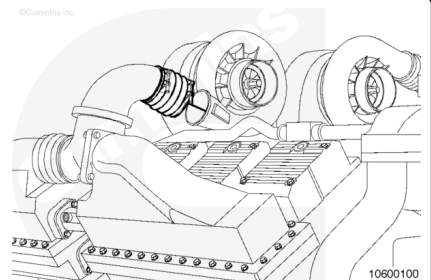
Remove the air intake piping from the turbocharger.



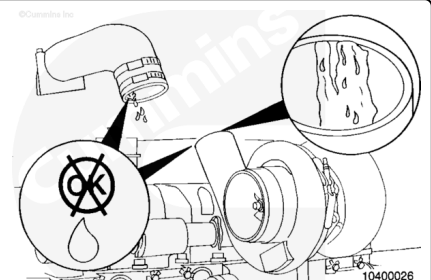
Remove the hose and two clamps between the air crossover and the turbocharger.

Remove the clamp between the air crossover and the aftercoolers or intercoolers.

Remove the air crossover.

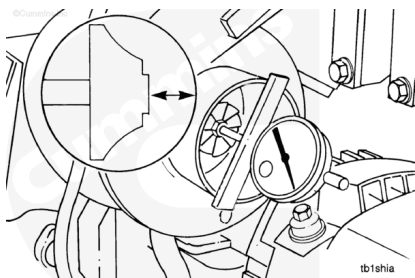


Inspect the compressor discharge and air crossover piping for oil.

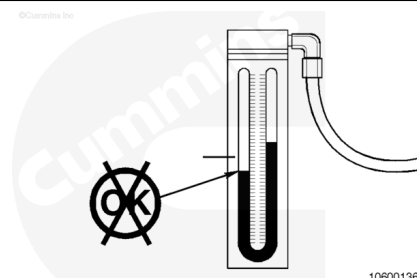


If oil is in the compressor discharge or crossover piping, check the axial clearance of the turbocharger.

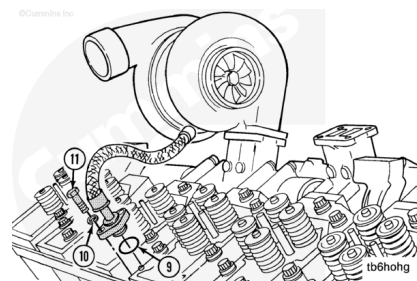
Check the radial clearance of the turbocharger.



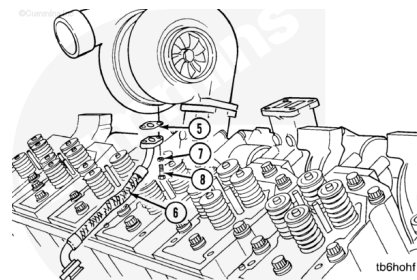
Check for intake restriction. Refer to Procedure 010-031 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-031.html>)



Disconnect the oil drain hose from the cylinder block.
Remove and discard the o-ring seal (9).



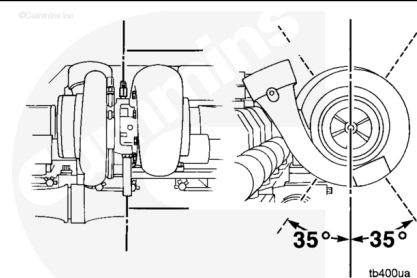
Disconnect the oil drain hose from the turbocharger.
Remove any support clamps and the hose.
Remove and discard the o-ring gasket (5).
Clear any restrictions found in the hose.
Discard the hose if it is cracked, frayed, or will **not** bend without cracking.



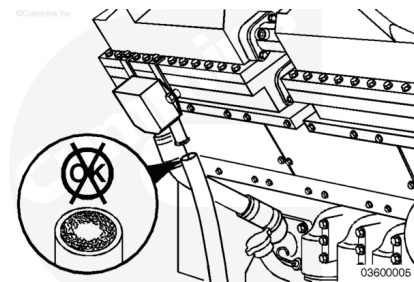
Check the angle of the drain tube.

The angle of the tube **must** be within 35 degrees of vertical. Adjust the turbocharger, if necessary. Use the following procedure for the high-pressure turbochargers. Refer to Procedure 010-034 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>)

Use the following procedure for the low-pressure turbochargers. Refer to Procedure 010-035 in Section 10 (</qs3/pubsys2/xml/en/procedures/56/56-010-035-tr.html>).



If the drain tube is free of restrictions and at the correct angle, check the crankcase breathers and tubes to make sure they are **not** plugged or restricted.

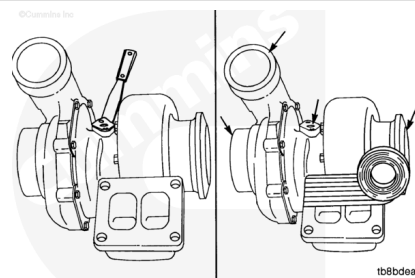


If these checks do **not** reveal the problem, measure the crankcase pressure (blowby). Refer to Procedure 014-005 in Section 14. (/qs3/pubsys2/xml/en/procedures/56/56-014-005-tr.html)



⚠ CAUTION ⚠

Cover all openings of the turbocharger to prevent steam or solvent from entering the oil cavities. Steam or solvent in the oil cavities can cause premature wear and damage.

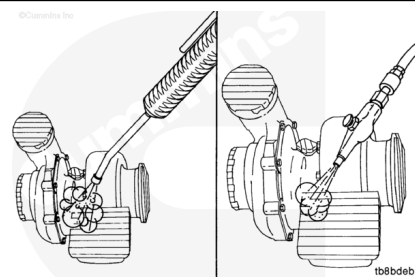


Clean gasket areas of the turbocharger.

Cover all openings of the turbocharger.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

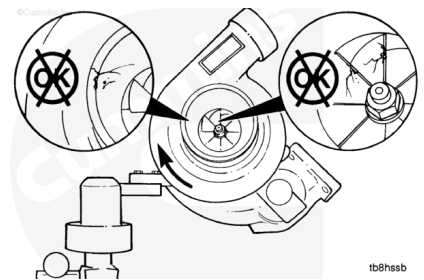
Clean the exterior of the turbocharger with solvent or steam.

Dry the turbocharger with compressed air.

Check the turbine and compressor impellers for fretting.

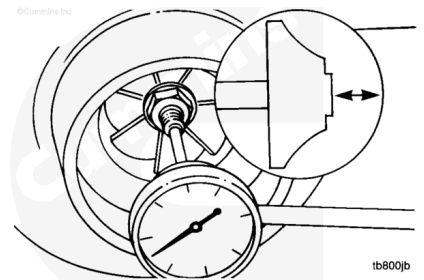
Check the vanes for cracks or other damage.

Repair or replace the turbocharger if the vanes are damaged.



Measure

Use a depth gauge, Part Number ST-537, or a dial indicator to measure the axial motion (end-to-end).



Compressor Impeller Axial Endplay			
	mm		in
Holset® HX82/HX83/HE851 Turbocharger/HX80	0.025	MIN	0.001
	0.152	MAX	0.006
Holset® HX60 Turbocharger	0.051	MIN	0.002
	0.152	MAX	0.006
Holset® HX55 Turbocharger	0.025	MIN	0.001
	0.127	MAX	0.005

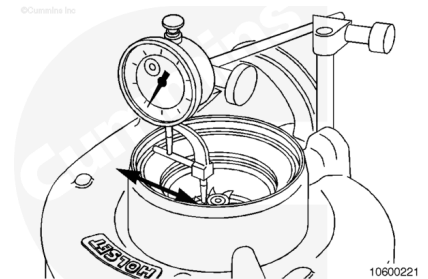
Holset® HE850	0.009	MIN	0.0003
	0.115	MAX	0.0045

Make sure the movement is within the MIN/MAX Total Indicator Reading (TIR) values shown above.

Note : If the axial clearance exceeds the specifications, the turbocharger must be replaced or rebuilt. Contact a Cummins® Authorized Repair Location.

Press the compressor impeller end of the shaft toward the housing.

Measure the radial clearance (side-to-side) at the compressor impeller nose using a dial indiactor.



Compressor Impeller Radial Endplay			
	mm		in
Holset® HX82/HX83/HE851 Turbocharg er/HX80	0.254	MIN	0.01
	0.787	MAX	0.031
Holset® HX60 Turbocharg er	0.406	MIN	0.016
	0.584	MAX	0.023
Holset® HX55 Turbocharg er	0.406	MIN	0.016
	0.610	MAX	0.024
Holset® HE850	0.062	MIN	0.0024
	0.848	MAX	0.0333

Make sure the movement is within the MIN/MAX TIR values shown above.

Note : If the radial clearance exceeds the specifications, the turbocharger must be replaced or

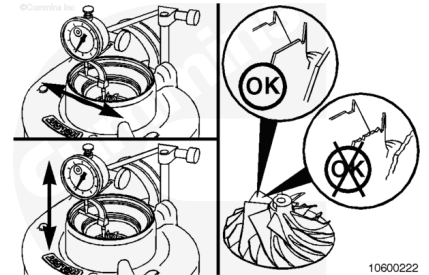
rebuilt. Contact a Cummins® Authorized Repair Location.

Disassemble

HX55 and HX60 Turbochargers

Before disassembly, check the turbine impeller and compressor impeller for damage.

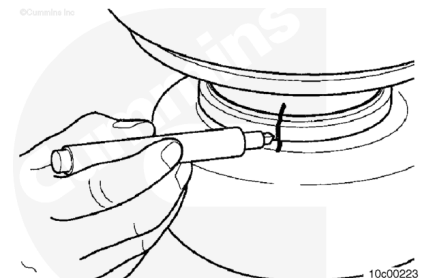
Measure the bearing radial movement and axial clearance to make sure the center housing rotating assembly is within specifications. Reference the Measure Compressor Impeller Endplay in the Clean and Inspect for Reuse section of the procedure.



Place the turbine outlet on a clean flat surface. Mark the turbine housing, bearing housing, V-band clamp, and turbine. This action assists in the reassembly process defining correct component orientation.

Loosen the locknut and V-band clamp. Discard the components.

Remove the compressor and turbine housings.

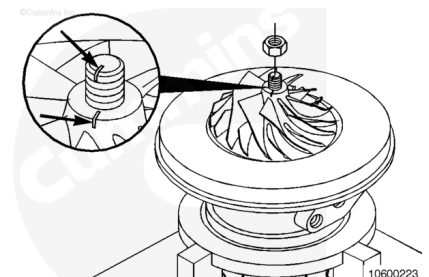


Always make sure the turbine wheel shaft and compressor impeller is marked before disassembly of the center housing rotating assembly.

If marks do **not** exist, scribe the shaft and compressor impeller before removing the compressor impeller locknut.

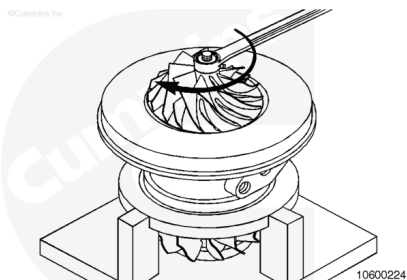
The cleaning process can remove indelible ink. A mark filed on the shaft and a line scribed on the compressor impeller nose is recommended.

Place the center housing rotating assembly on a suitable fixture with a 19 mm 12 point socket on the turbine impeller.

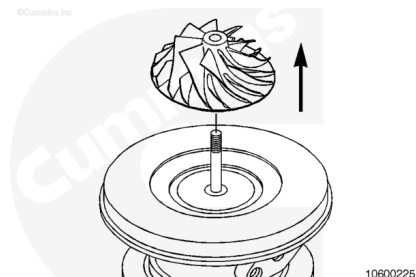


Use a 13 mm wrench to turn the compressor impeller locknut in a **clockwise** direction to remove it.

Remove the compressor impeller locknut.

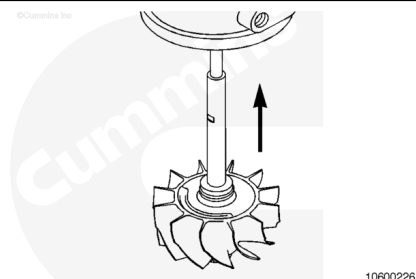


Remove the compressor impeller.



It is permissible to tap the protruding turbine shaft gently with a soft hammer if the split ring seals are stuck in the bore.

Slide the center housing off the turbine shaft.



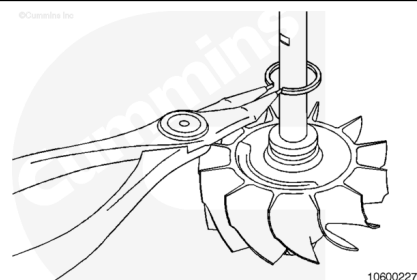
⚠ WARNING ⚠

To reduce the possibility of personal injury, always wear safety glasses when removing or installing a sealing ring.

Do **not** score the turbine wheel or shaft when removing the sealing rings.

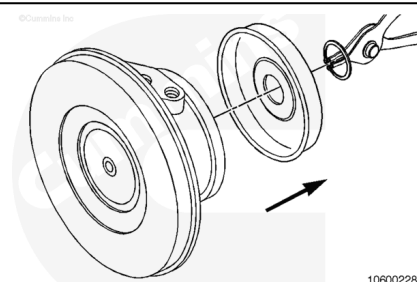
Carefully remove the twin split sealing rings from the turbine shaft.

Discard the sealing rings.

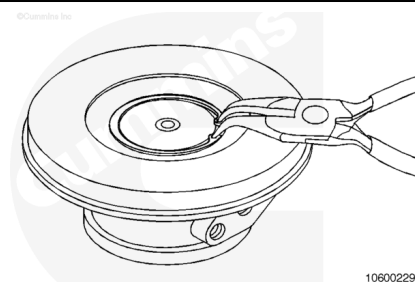


Use snap ring pliers to remove the heat shield snap ring.

Remove the heat shield.



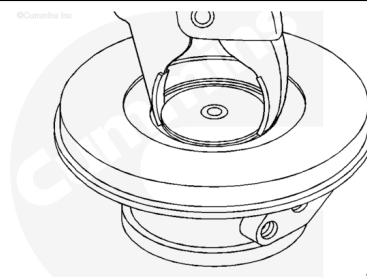
With the bearing housing on a clean flat surface, use snap ring pliers to remove the insert snap ring.



10600229

⚠ WARNING ⚠

Do not use excessive force when removing the oil seal plate assembly. Fragments can break off at high velocity. To reduce the possibility of personal injury, always wear safety glasses when removing the oil seal plate assembly.

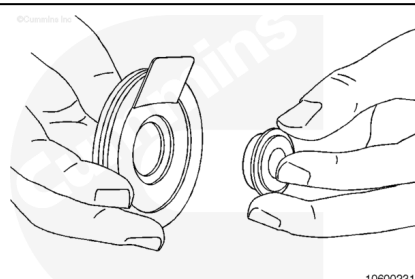


10600230

Use a pair of pliers to remove the oil seal plate assembly.

If the oil seal plate is stuck in place, it is permitted to use two screwdrivers positioned under the exposed lip to pry out the seal plate assembly.

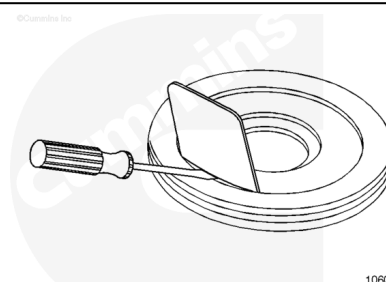
Remove the oil slinger from the oil seal plate.



10600231

Use a small screwdriver to release the oil baffle from the oil seal plate.

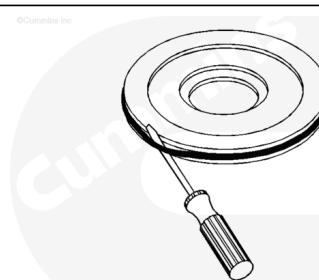
Discard the oil baffle.



10600232

Remove the o-ring seal from the oil seal plate.

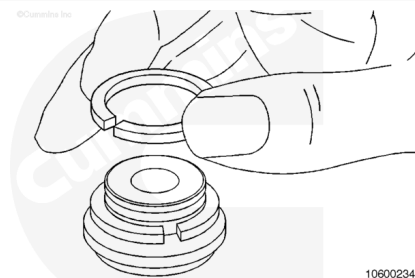
Discard the o-ring.



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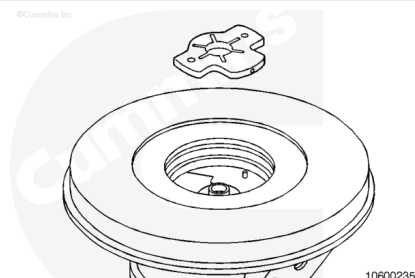
Remove the split ring seal(s) from the oil slinger.

Discard the split seal(s).

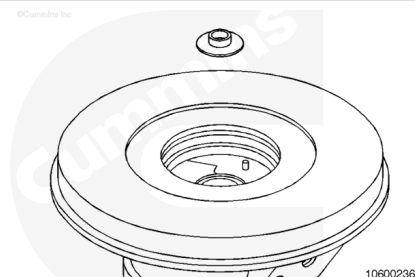


Remove the thrust bearing from the pin locations in the housing.

Discard the thrust bearing.



Remove the thrust collar.

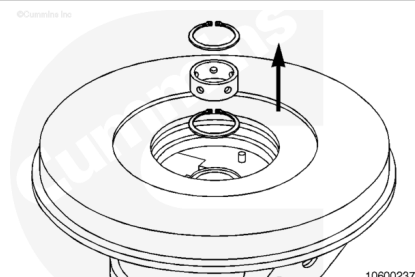


Use snap ring pliers to remove the snap ring from the compressor side bearing bore.

Remove the journal bearing.

Remove the lower snap ring.

Discard the snap rings and bearing.



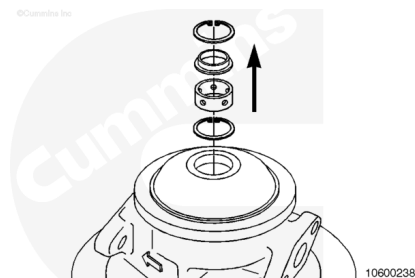
Turn the housing over.

Use snap ring pliers to remove the snap ring from the turbine side bearing bore.

Remove the journal bearing.

Remove the lower snap ring.

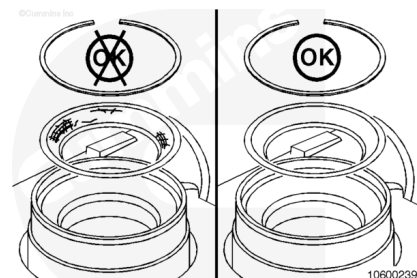
Discard the snap rings and bearing.



Certain turbochargers can have a pressed steel baffle fitted into the inlet section.

The baffle is **not** serviceable and **must** be replaced if damaged.

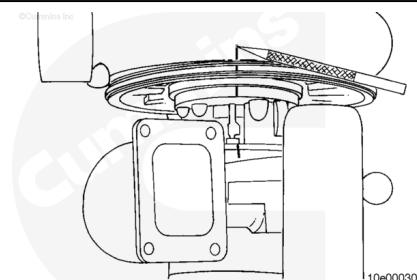
Remove the baffle steel retaining ring and baffle.



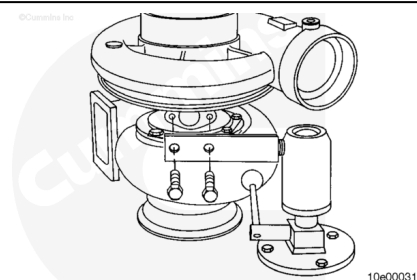
HX82 Turbocharger

Scribe a line between the turbocharger compressor housing and the center housing rotating assembly.

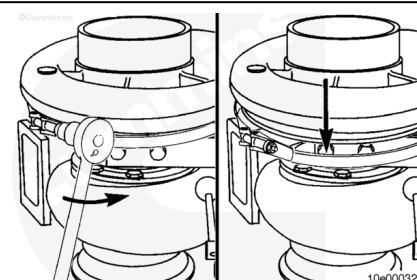
Scribe a line between the turbocharger turbine housing and the center housing rotating assembly.



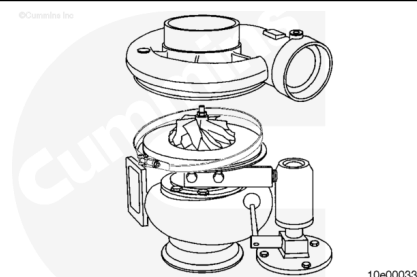
Place the turbocharger into a suitable vise.



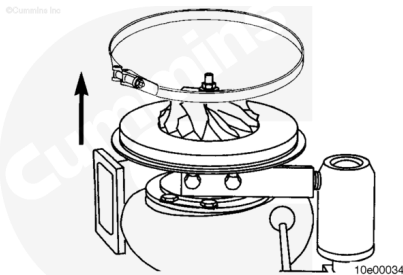
Loosen the compressor V-band clamp nut.



Remove the compressor housing.



Remove the compressor V-band clamp.

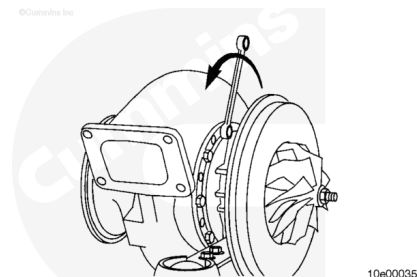


⚠ CAUTION ⚠

Use care when handling the turbocharger when the compressor housing is removed. The compressor impeller blades can be easily damaged.

Use a hammer and chisel to bend the lock tabs away from the capscrews.

Remove the capscrews and locking plates.



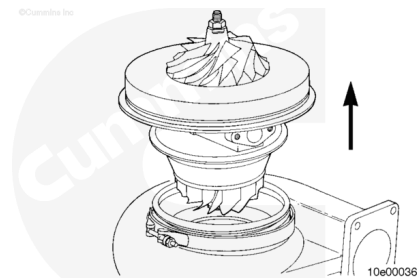
⚠ CAUTION ⚠

Never hit the compressor impeller. This can damage the axial thrust system and severely damage the blades of the compressor impeller.

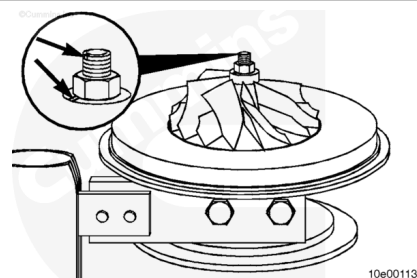
Rotate the center housing rotating assembly in the turbine housing.

If the center housing rotating assembly will **not** rotate in the turbine, tap the housing several times with a hammer to loosen the center housing rotating assembly.

Separate the center housing rotating assembly from the turbine housing.

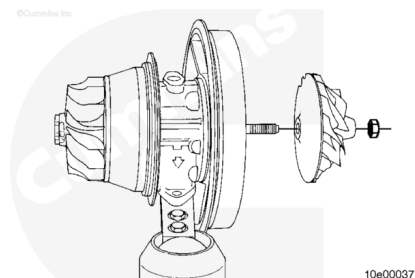


Scribe aligning marks on shaft and impeller for rebuild alignment. Failure to properly align the shaft and wheel can cause rotor assembly balance issues.



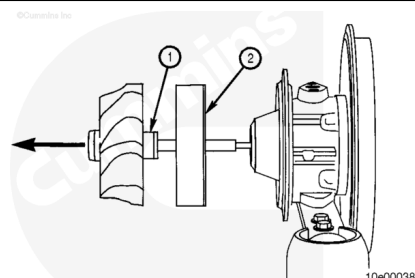
Remove the compressor impeller locknut.

Remove the compressor impeller.

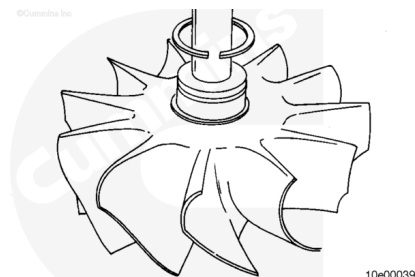


Use care when removing the turbine shaft so the turbine shaft is **not** scratched.

Remove the turbine shaft (1) and heat shield (2).

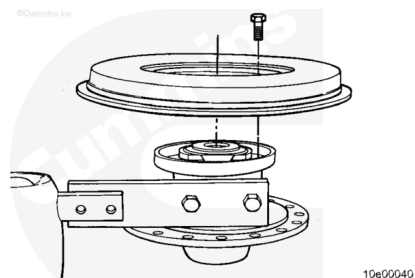


Remove the sealing ring on the turbine side of the shaft.

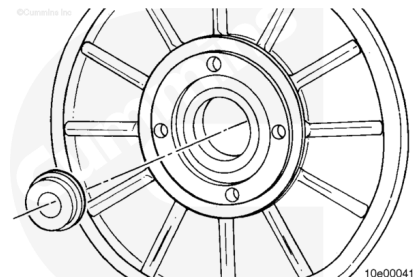


Remove the four diffuser mounting capscrews.

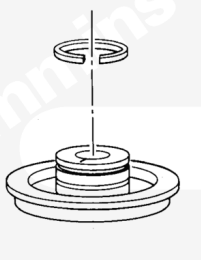
Remove the diffuser.



Remove the nozzle vane ring from the diffuser.



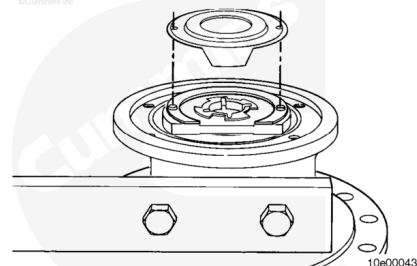
Remove the oil seal ring from the nozzle vane ring.



10e00042

Remove the oil partition and the o-ring.

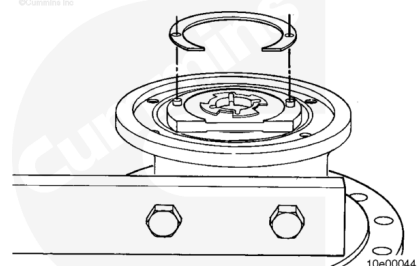
©Cummins Inc.



10e00043

Remove the axial clearance adjustment shim.

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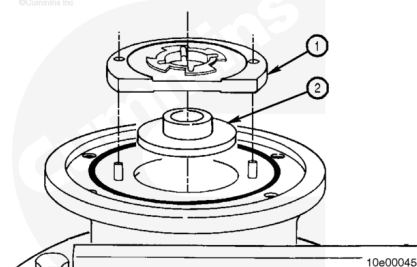


10e00044

Remove the axial thrust plate (1).

Remove the thrust ring (2).

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10e00045

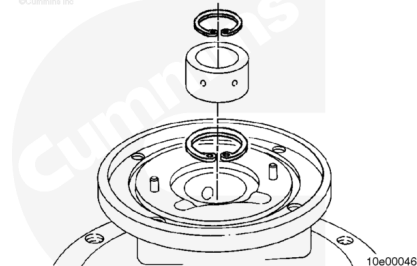
Use a pair of snap ring pliers to remove the upper snap ring.

Remove the journal bearing.

Remove the lower snap ring.

Turn the housing over and repeat the process.

©Cummins Inc.

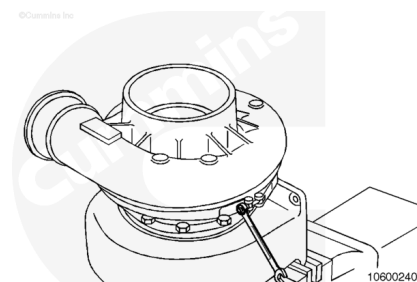


10e00046

HX83 Turbocharger

⚠ CAUTION ⚠

HX83 cores with titanium compressor wheels must be core balanced using specialist equipment. Reference a Holset® approved agent for overhaul and repair.



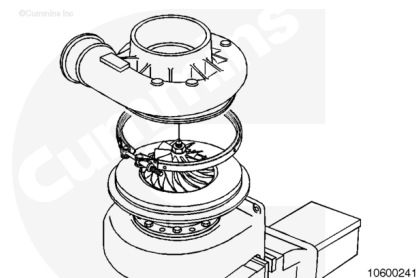
The center housing rotating assembly is replaced as a complete assembly for the HX83 turbocharger.

Place the turbocharger in an appropriate vise.

Loosen the compressor V-band locknut.

⚠ CAUTION ⚠

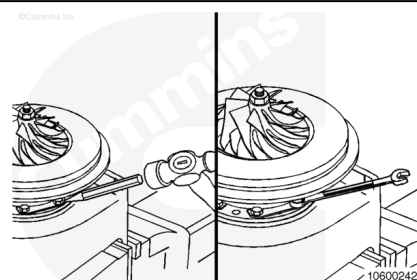
Use care when removing the compressor housing. The compressor impeller blades can be easily damaged when the compressor housing is removed.



Carefully remove the compressor housing and V-band clamp.

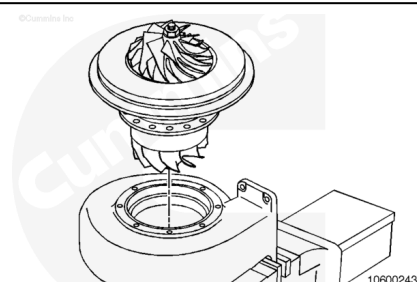
Use a hammer and a chisel to bend the lockplate tabs away from the turbine housing capscrews.

Remove and discard the turbine housing capscrews and lockplates.



⚠ CAUTION ⚠

Use care when separating the center housing rotating assembly from the turbine housing. The turbine blades can be easily damaged when the center housing rotating assembly is separated from the turbine housing.



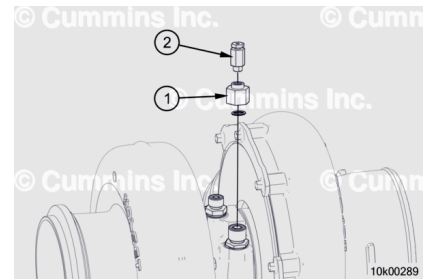
Separate the center housing rotating assembly from the turbine housing.

HE850 Turbocharger

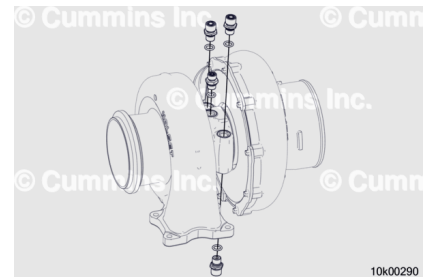
The Holset® HE850 turbocharger can **not** be serviced or repaired in the field. The turbocharger **must** be returned to a local Holset® service center for service or repair.

Remove coolant vent (2).

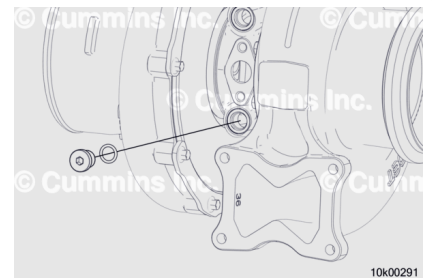
Remove reducing adapter (1). Remove and discard o-ring seal.



Remove male unions. Remove and discard o-ring seals.



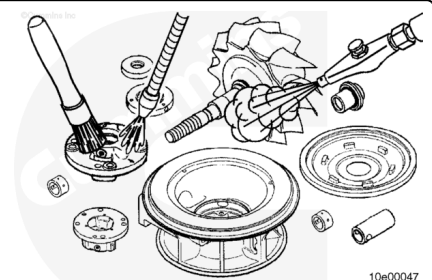
Remove straight thread o-ring plug. Remove and discard o-ring seal.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not bead blast an area of a component for a long period of time. Bead blasting areas for a long period of time can affect the component balance.

Soak all the components in a non-caustic metal cleaner to loosen deposits.

Never use a wire brush or a metal scraper on any turbocharger component.

A bristle brush can be used on aluminum components.

Dry the components with compressed air.

Bead blast the turbine housing if chemicals do **not** clean it.

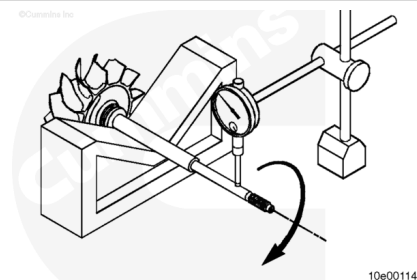
It can be necessary to bead blast the seal ring groove area of turbine wheel using a low grade medium shot.

If bead blasting is required for any component, the cleaning process **must** be repeated.

The surface adjacent to the turbine and compressor wheels on the stationary housings **must** be clean, smooth, and free of deposits.

⚠ CAUTION ⚠

Do not attempt to straighten a bent shaft. Premature turbocharger damage can occur.



Place the turbine shaft on a vee block.

Set a dial indicator on the shaft.

Rotate the turbine wheel and check for movement on the dial indicator.

Turbocharger Shaft Bend		
mm		in
0.015	MAX	0.0006

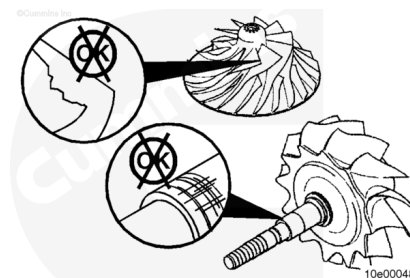
If the turbine shaft is **not** within specifications, the turbine shaft **must** be replaced.

Inspect the compressor and turbine impellers for nicks and cracks. Inspect the blade tips for wear.

If an impeller is damaged, it **must** be replaced.

Inspect the turbine shaft for wear or damage.

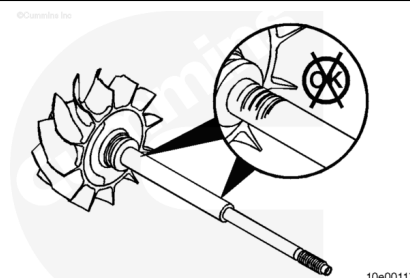
If turbine shaft is damaged, it **must** be replaced.



Inspect the bearing journals for excessive scratches and wear.

Minor scratches are acceptable.

Measure the turbine bearing journal area.



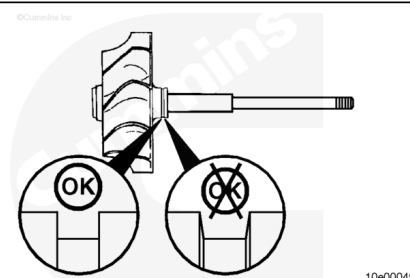
Turbocharger Turbine Shaft Journals

	mm		in
Holset® HX60 Turbocharg er	14.25	MIN	0.561
Holset® HX82 Turbocharg er	15.965	MIN	0.628
Holset® HX55 Turbocharg er	12.19	MIN	0.48

If the turbocharger turbine shaft journals are **not** within specifications, the turbocharger turbine shaft **must** be replaced.

Inspect the sealing ring groove for wear. The walls of the groove **must** be perpendicular with the shaft center line.

Measure the sealing ring groove.



Turbocharger Turbine Shaft Sealing Ring Groove Width

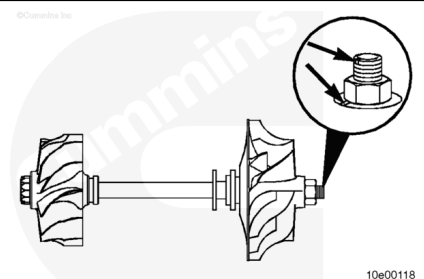
	mm		in
Holset® HX60 Turbocharg er	1.90	MIN	0.075

Holset® HX82 Turbocharg er	1.664	MIN	0.067
Holset® HX55 Single Ring	1.69	MIN	0.067
Holset® HX55 Dual Ring	3.30	MIN	0.130

If the turbocharger turbine shaft sealing ring groove width is **not** within specifications, the turbocharger turbine shaft **must** be replaced.

⚠ CAUTION ⚠

Make sure the compressor and turbine impellers are individually balanced. If not balanced, the impellers can cause excessive vibration and premature turbocharger malfunction.



Check the balance of the turbine shaft assembly to make sure it is within specifications.

Holset® HX60 and HX 82 Turbine Shaft Balance Limits

Turbine Impeller End	Compressor Impeller End
7.2 gmm [0.010 oz ins]	12.3 gmm [0.017 oz ins]

Holset® HX55 Turbine Shaft Balance Limits

Turbine Impeller End	Compressor Impeller End
2.2 gmm [0.003 oz ins]	4.4 gmm [0.006 oz ins]

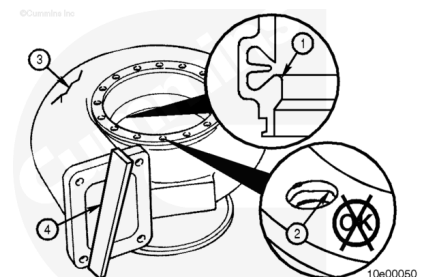
Mark the compressor impeller and shaft with a marker to aid in alignment during assembly.

Inspect the turbine housing for signs of turbine impeller rub (1).

If any signs of turbine impeller rubbing are found, the turbine housing **must** be replaced.

Inspect the tapped holes for damaged threads (2).

Repair damaged threads or replace the turbine housing.



Check for signs of cracking or flaking caused by possible overheating (3).

Replace the turbine housing if it is cracked.

Check the flange for flatness.

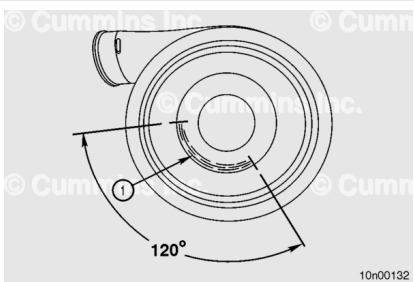
Turbocharger Turbine Housing Flange Flatness		
mm		in
0.010	MIN	0.004

If the turbocharger turbine housing flange is **not** within specifications, the turbocharger turbine housing **must** be repaired or replaced.

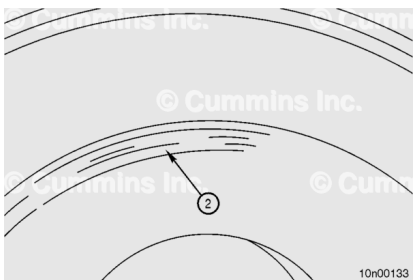
Turbochargers for highly rated or extreme duty cycle engines will, in some cases, employ a titanium compressor wheel with a cast iron compressor cover.

These materials have different expansion rates, and dependent upon operator use, the compressor wheel can thermally expand at an elevated rate compared to the compressor cover. This has potential to lead to light contact between the compressor cover and compressor wheel. This contact can leave a cosmetic witness mark at the outer diameter of the compressor wheel blade tips. The marking can be evident on both components, the compressor cover and the compressor wheel, or just on one component.

The level of contact is variable and depends upon a number of factors. Approximately 120 degrees of contact (1):

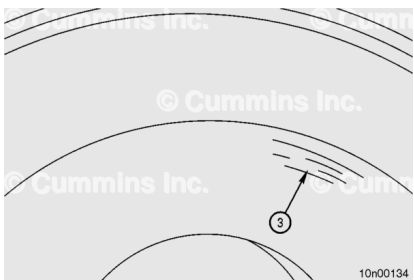


A close up view (2) of the contact area:



Approximately 5 degrees of contact (3):

If this contact marking is observed, the turbocharger bearing clearances **must** be measured and the compressor V-band torque verified. Provided the turbocharger bearing clearances are within specification limits and the compressor cover V-



band torque is correct, this type of damage is cosmetic and will **not** cause an issue with the performance of the turbocharger. Under this circumstance, the turbocharger does **not** need to be replaced.

If the bearing clearances are found out of specification or the compressor V-band is loose, the turbocharger **must** be replaced.

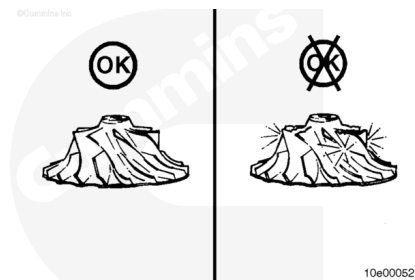
This guidance is only relevant to turbochargers fitted with a titanium compressor wheel and a cast iron compressor cover.

For all other turbochargers apart from those mentioned above, check the compressor housing for signs of the compressor impeller rubbing.

If any signs of the compressor impeller rubbing are found, replace the compressor housing.

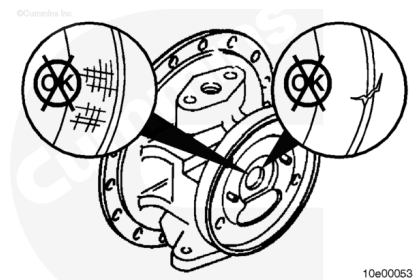
Inspect the compressor impeller for bent or cracked blades or blades worn at the end.

Replace the compressor impeller if any damage or wear is found.



Inspect the center housing for signs of wear or cracking in the bearing bore and retaining groove area.

Replace the center housing if it is damaged or worn.

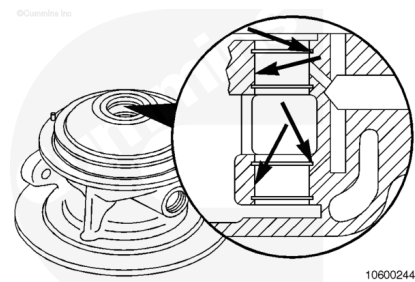


HX55 and HX60 Turbochargers

Inspect the journal bearing bores for wear and score marks.

Replace the center housing if scratching is severe.

Measure the center housing bearing bore.



Holset® HX55 and HX60 Turbochargers Center Housing Journal Bearing Bore		
mm		in
22.27	MAX	0.877

If the turbocharger center housing journal bearing bore is **not** within specifications, the turbocharger center housing **must** be replaced.

Make sure the groove locations for the journal bearing snap rings are free from deposits.

Measure the center housing journal bearing snap ring groove width.

Holset® HX55 and HX60 Turbochargers Center Housing Journal Bearing Snap Ring Groove Width		
mm		in
1.25	MAX	0.049

If the turbocharger center housing journal bearing snap ring groove width is **not** within specifications, the turbocharger center housing **must** be replaced.

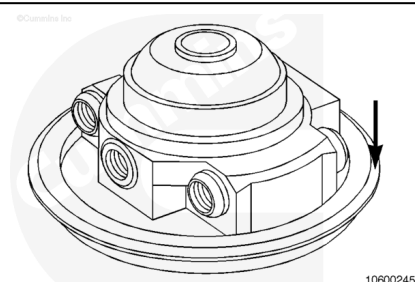
Make sure the oil cavity is free from carbon and debris that can result in failure.

Make sure all oil port threads are clean and checked with the appropriate thread gauge.

Oil Port Threads	
Oil Flanges	Oil Adapter
M8 x 1.25	M16 x 1.5

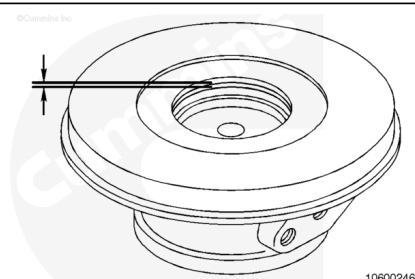
Make sure all the coolant port threads are clean.

Check the threads with a M18 x 1.5 thread gauge.



Measure the oil seal plate snap ring groove width.

Turbocharger Oil Seal Plate Retaining Ring Groove Width		
mm		in
2.0	MAX	0.079

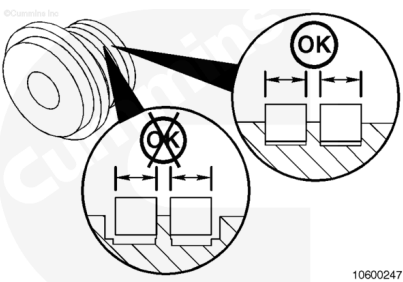


If the turbocharger oil seal plate retaining ring groove width is **not** within specifications, the center housing **must** be replaced.

Inspect the oil slinger sealing ring grooves for scoring or other damage.

Replace the oil slinger if the sealing ring grooves are damaged.

Measure the oil slinger sealing ring groove width.



Turbocharger Oil Slinger Sealing Ring Groove Width

mm		in
1.70	MAX	0.067

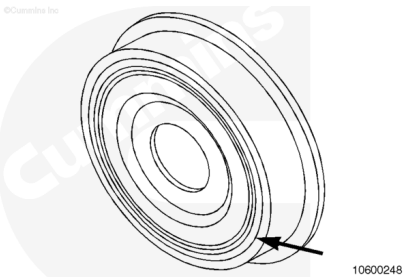
If the turbocharger oil slinger sealing ring groove width is **not** within specifications, the oil slinger **must** be replaced.

Check for signs of rubbing and scoring on the thrust surface of the oil slinger.

Replace the oil slinger if damaged.

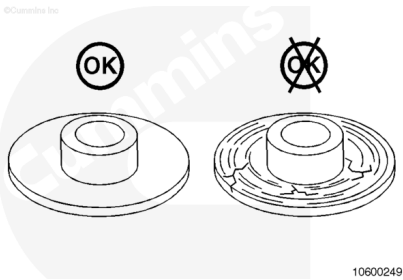
Check the heat shield for signs of distortion, rubbing, or cracking.

Replace the heat shield if damaged.



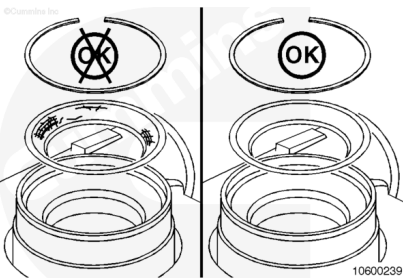
Check the thrust collar for cracks and scoring on the thrust face.

Replace the thrust collar if it is cracked or scored.



Check the baffle for damage.

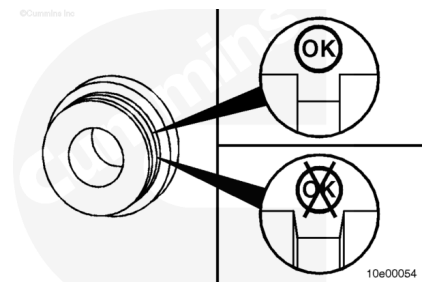
The baffle is **not** serviceable and **must** be replaced if damaged.



Check the oil slinger groove for wear and signs of cracking.

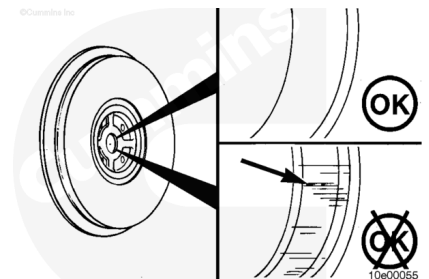
Check the oil slinger groove walls for signs of friction.

Replace the oil slinger and seal ring if damage is found.



Check the diffuser bore for excessive wear or damage.

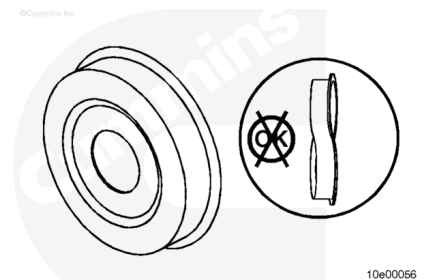
Replace the diffuser plate if damaged or excessively worn.



Make sure the heat shield is **not** warped.

Check the heat shield for signs of rubbing.

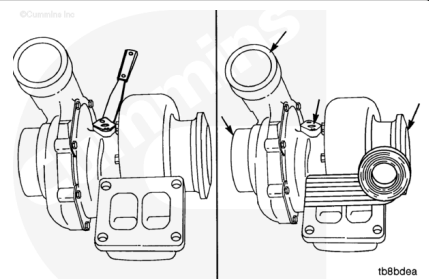
Replace the heat shield if warped or damaged.



HE850 Turbocharger

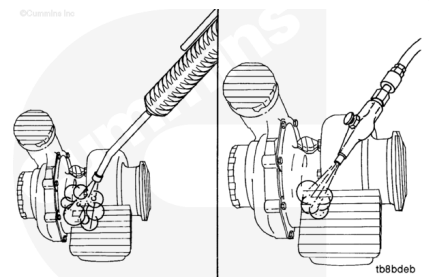
⚠ CAUTION ⚠

Tape or plug all openings to prevent solvent or steam from entering the oil cavities in the turbocharger.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

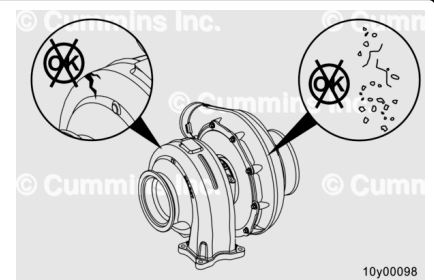
Clean:

- Exterior of turbocharger
- Male unions
- Straight thread o-ring plug
- Coolant vent
- Reducing adapter

Use solvent or steam. Dry with compressed air.

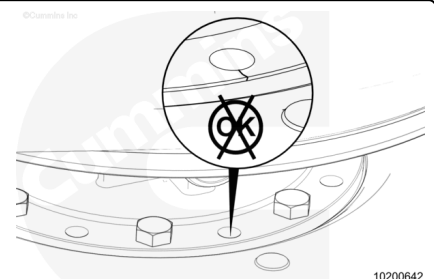
Inspect for cracks in turbine housing and compressor housing.

If through cracks are found in outer walls, replace turbocharger.



Inspect turbocharger diffuser plate and bearing housing for cracks without removing mounting hardware.

If a crack is found, perform further inspection in next step.



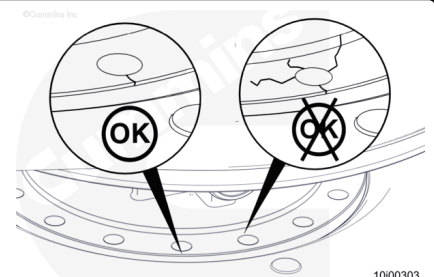
Only perform step if crack found during previous step.

Inspect turbocharger diffuser plate.

Replace turbocharger if cracks are identified at the diffuser plate.

Inspect turbocharger bearing housing.

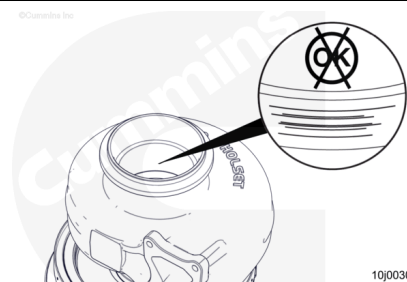
Cracks running from center of flange holes towards outer diameter of flange are acceptable on the bearing housing.



Loosely install new capscrews and locking plates. Capscrews will be tightened in Install section below.

Inspect compressor housing or turbine housing for signs of impeller wheel or turbine wheel rub.

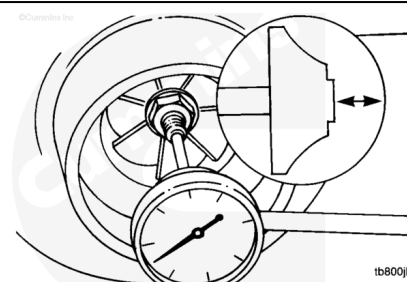
If any signs of impeller wheel or turbine wheel rubbing are found, replace turbocharger.



10j00304

Use a dial depth gauge, Part Number ST-537, or a dial indicator to measure the axial clearance.

Holset® HE850 Compressor Axial Clearance		
mm		in
0.009	MIN	0.0003
0.115	MAX	0.0045



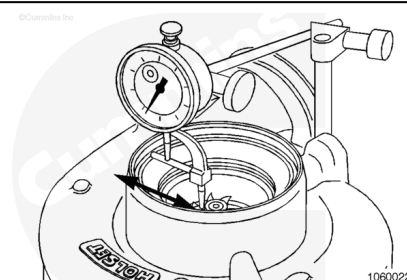
tb800jb

Make sure the movement is within the minimum/maximum total indicator reading values shown above.

If the end clearance exceeds the specifications, the turbocharger **must** be replaced or rebuilt. Contact a Cummins® Authorized Repair Location.

Measure the radial clearance (side-to-side) at the compressor nose, using a dial gauge.

Holset® HE850 Compressor Radial Clearance		
mm		in
0.062	MIN	0.0024
0.848	MAX	0.0333



10600221

Make sure the movement is within the minimum/maximum total indicator reading values shown above.

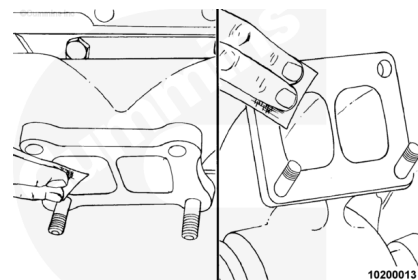
If the clearance exceeds the specifications, the turbocharger **must** be replaced or rebuilt. Contact a Cummins® Authorized Repair Location.

Note : All openings on the turbocharger, including the turbine inlet connection, must be plugged during flange cleaning.

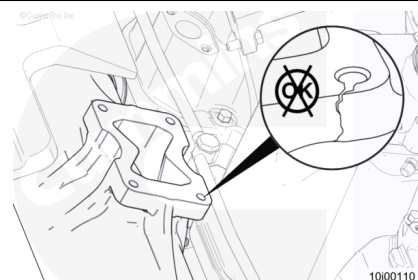
Clean turbine housing and exhaust manifold mating surfaces. Use Scotch-Brite™ 7448 abrasive pad.

After abrasive cleaning, wipe debris from both surfaces. Use clean shop towel.

The surface under the mounting nuts must be free of dirt, rust, or any other debris, before applying anti-seize compound, Part Number 3824397, or equivalent.



Inspect exhaust manifold outlet flange. Refer to Procedure 011-007 (/qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html) in Section 11.



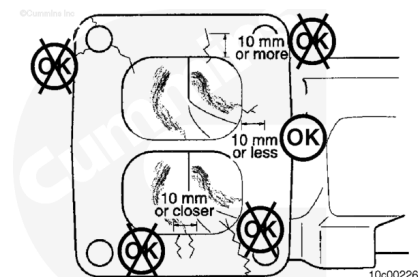
Inspect turbine housing inlet flange.

Replace turbocharger if:

- Cracked
- Pitted
- Otherwise damaged

Inspect turbine housing inlet flange flatness. It **must** be within 0.1 mm [0.004 in] to be acceptable for reuse.

If turbine housing inlet flange flatness is **not** within specifications, replace turbocharger.

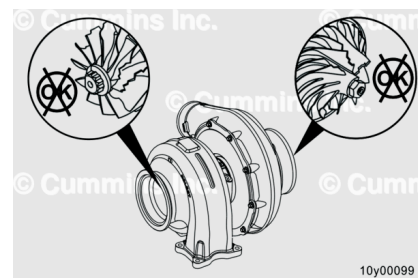


Note : If damage is found on either the turbine or compressor wheels, the turbocharger **must** be replaced or rebuilt. Contact a Cummins® Authorized Repair Location.

Inspect compressor wheel and turbine wheel.

Replace turbocharger if:

- Nicked
- Cracked
- Worn

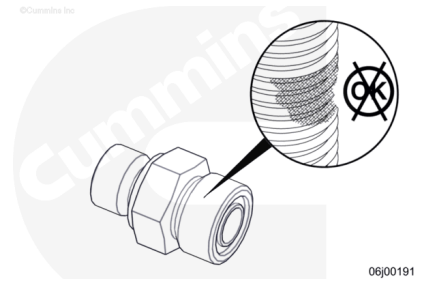


Inspect male unions.

Replace component if:

- Cracked

- Corroded
- Otherwise damaged

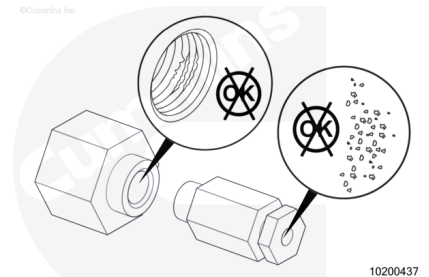


Inspect:

- Coolant vent
- Reducing adapter

Replace component if:

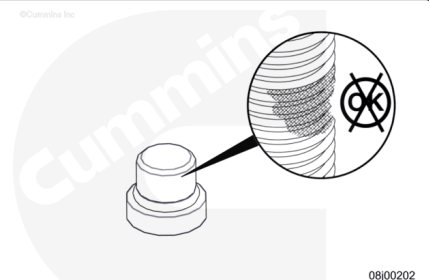
- Cracked
- Corroded
- Otherwise damaged



Inspect straight thread o-ring plug.

Replace plug if:

- Corroded
- Otherwise damaged



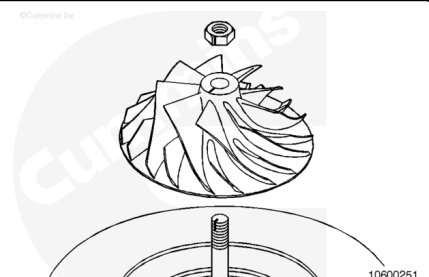
Assemble

HX55 and HX60 Turbochargers

The turbocharger manufacturer can use a separate part balancing process. The turbine shaft balance **must always** be checked when rebuilding the turbocharger.

The turbocharger turbine shaft will have marks on the end of the turbine shaft and the compressor impeller nose.

The thrust collar and oil slinger will **not** have marks.

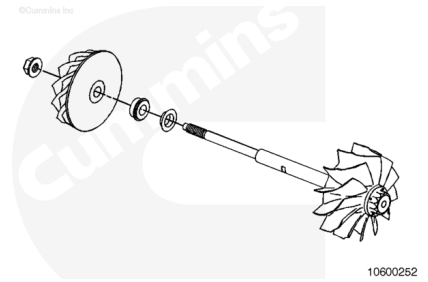


Only use a genuine Holset® compressor impeller and a turbine shaft which has been individually check balanced for long life and quiet operation.

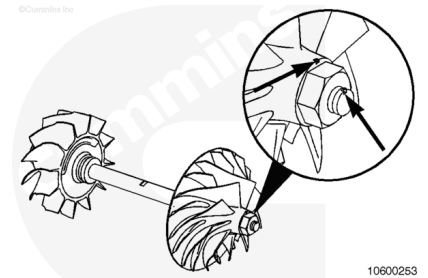
Components that are included in the turbine shaft balance are:

- Turbine wheel and shaft
- Thrust collar
- Oil slinger
- Locknut.

Balance is achieved by rotating the components in a purpose-built balance rig and in compliance with the balance rig procedure.



Always make sure the balance marks on the turbine shaft and compressor impeller are in alignment when rebuilding the turbocharger.

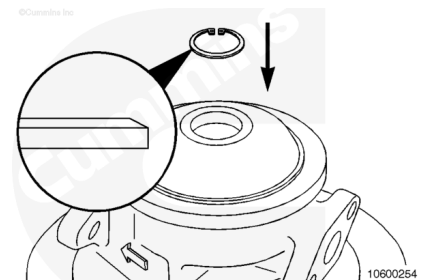


⚠ WARNING ⚠

To reduce the possibility of personal injury, always wear safety glasses when removing or installing a snap ring.

⚠ CAUTION ⚠

Make sure the snap ring is fitted into the center housing with the beveled edge facing the journal bearing. Premature turbocharger damage will result from incorrect installation.

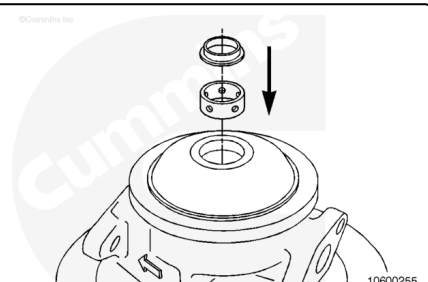


Place the center housing on a clean flat surface with the compressor mating face down.

Use snap ring pliers to install the turbine side inner journal bearing snap ring with the beveled edge facing the bearing.

Lubricate the journal bearing with clean engine oil and install it into the center housing.

Lubricate the oil control sleeve with clean engine oil and install it into the center housing.

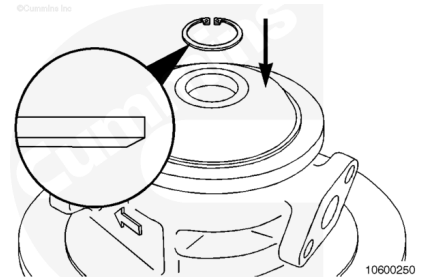


⚠ WARNING ⚠

To reduce the possibility of personal injury, always wear safety glasses when removing or installing a snap ring.

Use snap ring pliers to install the snap ring with the beveled edge facing the journal bearing.

Turn the center housing over and repeat the journal bearing installation process for the compressor side.

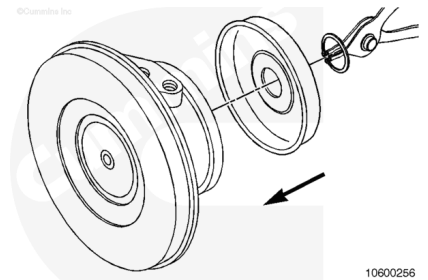


⚠ WARNING ⚠

To reduce the possibility of personal injury, always wear safety glasses when removing or installing a snap ring.

Install the heat shield.

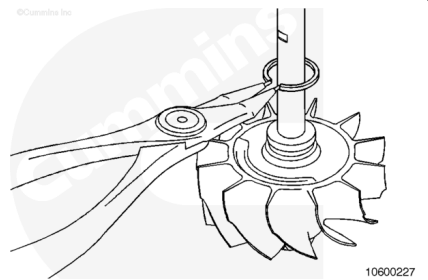
Use snap ring pliers to install the heat shield snap ring.



⚠ WARNING ⚠

To reduce the possibility of personal injury, always wear safety glasses when removing or installing a sealing ring.

Use ring expander pliers to install the new split ring seals into the grooves on the turbine shaft.



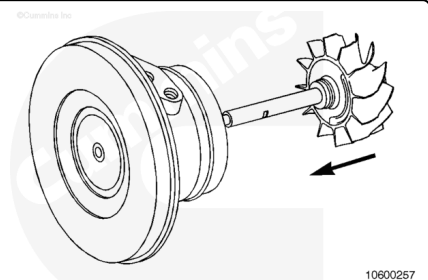
Align the split ring seal so the gap in the split rings is 180 degrees apart.

Lubricate the shaft with clean engine oil.

Insert the shaft into the center housing journal bearings.

A slight rotation of the wheel will assist in locating the split ring seals.

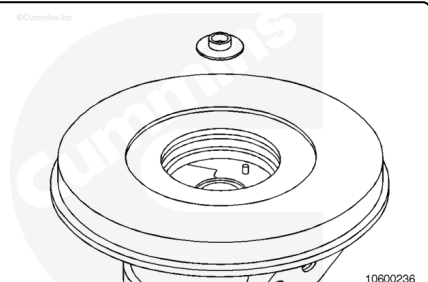
Make sure the turbine shaft rotates freely.



Support the turbine impeller in a suitable fixture.

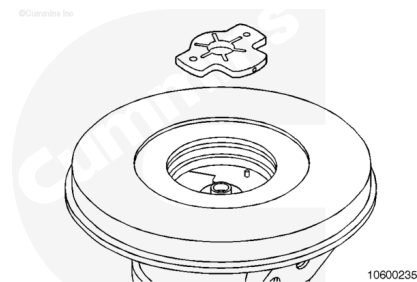
Make sure all thrust contact faces are clean.

Install the thrust collar.

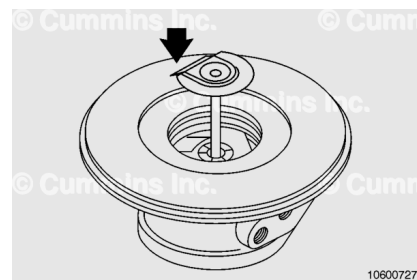


Apply clean engine oil to the installed thrust collar.

Install the thrust bearing onto its locating pins in the center housing.

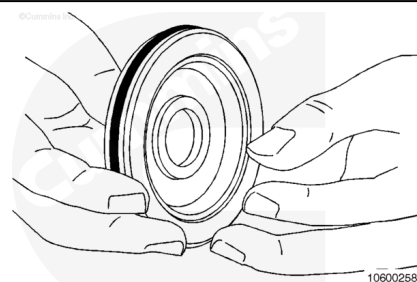


Install the oil baffle into the oil seal plate.

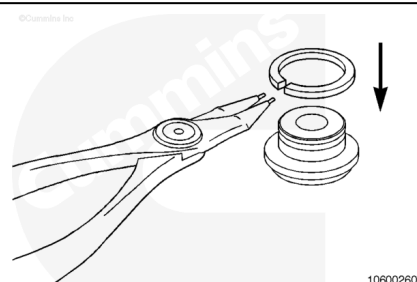


Apply a light coat of clean engine oil onto the oil seal plate.

Install a new o-ring onto the oil seal plate.

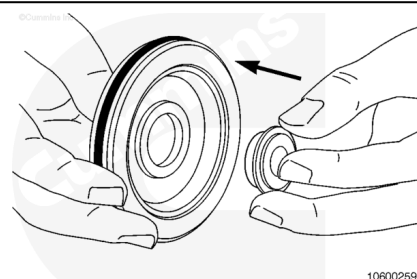


Install new split ring seals onto the oil slinger.

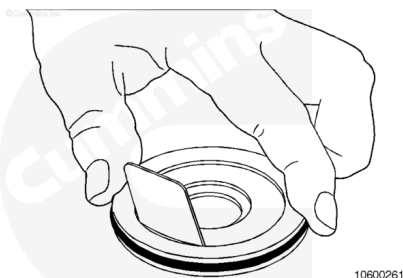


Lubricate the oil slinger with clean engine oil.

Insert the oil slinger into the oil seal plate.



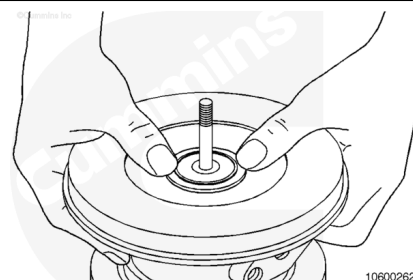
Press fit the oil baffle into the oil seal plate.



10600261

Place the center housing rotating assembly on the fixture supporting the housing.

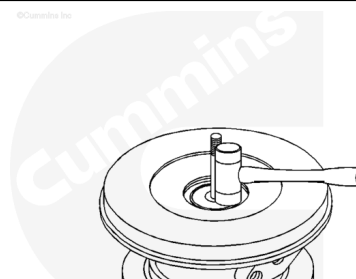
Install the oil seal plate into the center housing rotating assembly.



10600262

Tap the oil seal plate with a soft hammer to seat it into position.

Make sure the o-ring was **not** cut during the seal plate insertion.



10600263

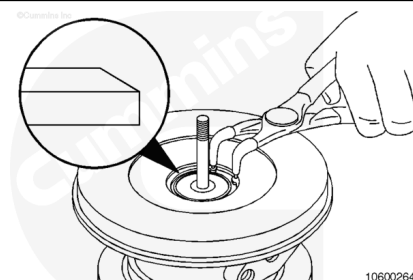
⚠ WARNING ⚠

To reduce the possibility of personal injury, always wear safety glasses when removing or installing a snap ring.

Place a 19 mm, 12 point socket onto the turbine impeller and set the center housing rotating assembly on a fixture supporting the socket.

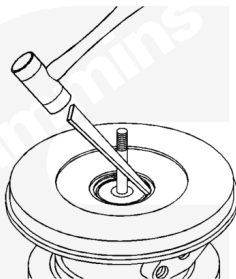
Use snap ring pliers to install the snap ring, with the chamfered side facing up, into the center housing rotating assembly.

It can be necessary to lift the center housing rotating assembly to fully expose the snap ring groove.



10600264

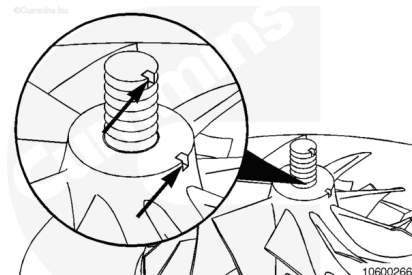
Use a soft hammer and drift to make sure the snap ring is properly installed.



10600265

Align the balance or scribe marks on the compressor impeller and turbine shaft.

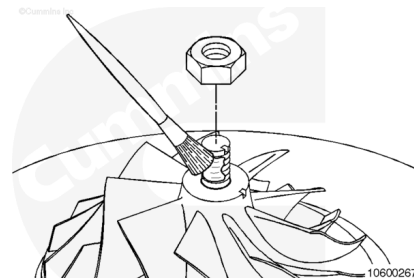
Install the compressor impeller onto the turbine shaft.



10600266

Apply a coat of anti-seize compound to the compressor impeller locknut face and turbine shaft threads to avoid loss of alignment during installation of the locknut.

Thread the nut onto the shaft in a **counterclockwise** direction.



10600267

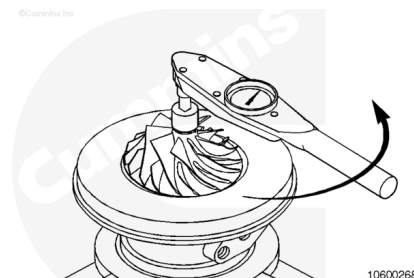
Tighten the compressor impeller locknut.

Torque Value:

Holset® HX60 41 n•m [30 ft-lb]

Torque Value:

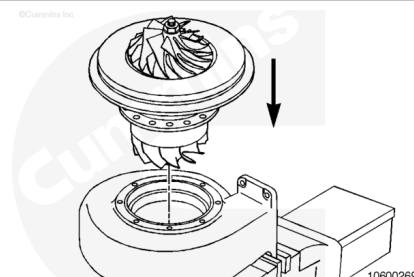
Holset® HX55 17 n•m [150 in-lb]



10600268

⚠ CAUTION ⚠

Use care when installing the center housing rotating assembly into the turbine housing. The turbine wheel blades can be easily damaged.



10600269

Set the turbine housing on a clean flat surface.

Place the V-band clamp into position loosely on the turbine housing.

Carefully slide the center housing rotating assembly into the turbine housing.

Align the marks on the turbine housing to the center housing rotating assembly.

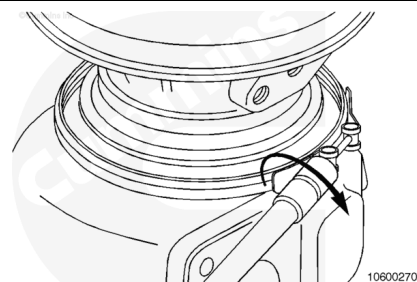
Position the turbine V-band clamp on the center housing rotating assembly and turbine housing.

Install a new locknut and tighten.

Torque Value:

Holset® HX55 and HX60 9 n•m [80 in-lb]

Make sure the turbine shaft rotates freely.



10600270

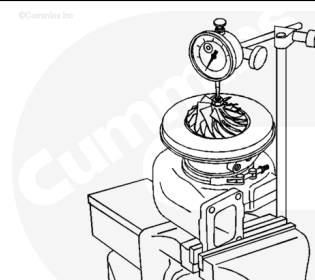
Check the turbocharger shaft end clearance.

Holset® HX60 Turbocharger Shaft End Clearance

	mm		in
Compress or Impeller	0.051	MIN	0.002
	0.152	MAX	0.006

Holset® HX55 Turbocharger Shaft End Clearance

	mm		in
Compress or Impeller	0.025	MIN	0.001
	0.127	MAX	0.005



10600271

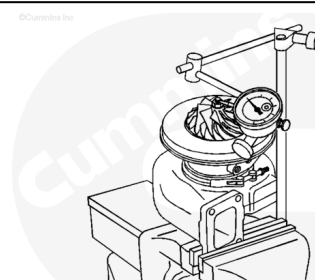
Check the turbocharger radial bearing clearance.

Holset® HX60 Turbocharger Radial Bearing Clearance

mm		in
0.406	MIN	0.016
0.584	MAX	0.023

Holset® HX55 Turbocharger Radial Bearing Clearance

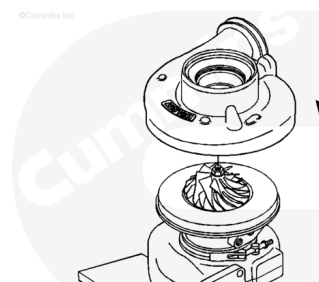
mm		in
0.406	MIN	0.016
0.610	MAX	0.024



10600272

⚠ CAUTION ⚠

Use care when installing the compressor housing onto the turbocharger. The compressor impeller can be easily damaged.



10600273

Place the turbocharger turbine housing on a flat clean surface.

Place the compressor housing V-band clamp loosely onto the center housing rotating assembly.

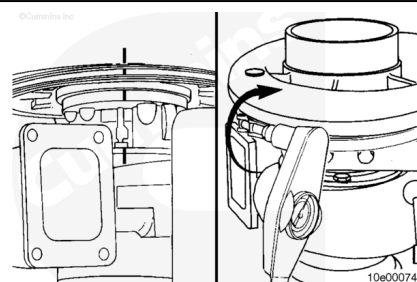
Carefully place the compressor housing over the compressor impeller onto the center housing rotating assembly.

Align the marks on the compressor housing to the marks on the center housing rotating assembly.

Position a new V-band clamp on the compressor housing and center housing rotating assembly.

Install a new nut and tighten.

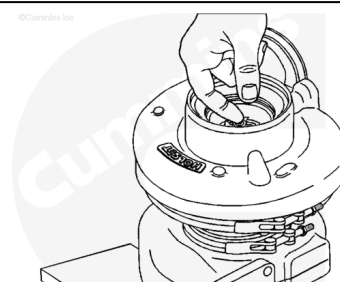
Torque Value: 9 n·m [80 in-lb]



10600074

Make sure the turbine shaft rotates freely.

Note : Some turbochargers have a coating applied to the internal surface of the turbine housing. This special coating is abraded by the turbine impeller and can affect turbocharger rotation until its unique form is fully formed.



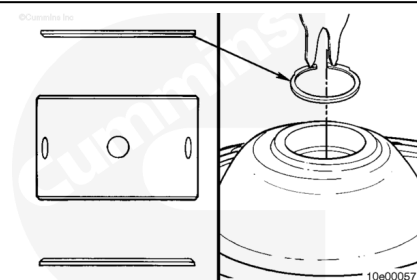
10600275

HX82 Turbocharger

⚠ CAUTION ⚠

The snap ring must be installed with the chamfer side of the snap ring facing the bearing. Premature damage can occur if the snap rings are installed incorrectly.

Insert the inner snap ring, with the chamfer side facing the bearing, into the turbine side of the center housing.

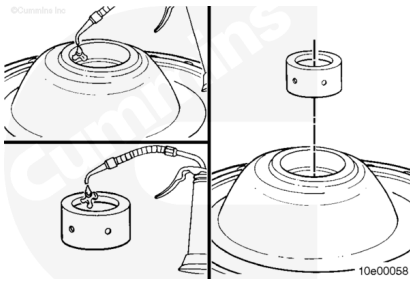


10600057

Lubricate the center rotating housing turbine side bearing shell.

Lubricate the inside diameter of the bearing.

Place the bearing into the center housing.

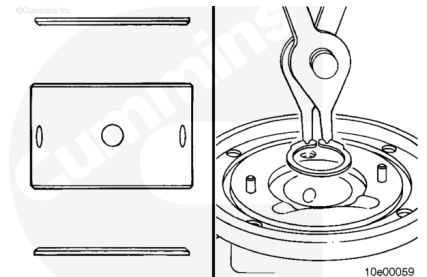


⚠ CAUTION ⚠

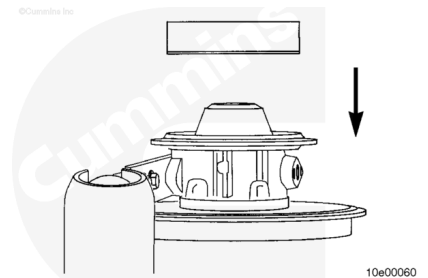
The snap ring must be installed with the chamfer side of the snap ring facing the bearing. Premature damage can occur if the snap rings are installed incorrectly.

Insert the outer snap ring, with the chamfer side facing the bearing, into the turbine side of the center housing.

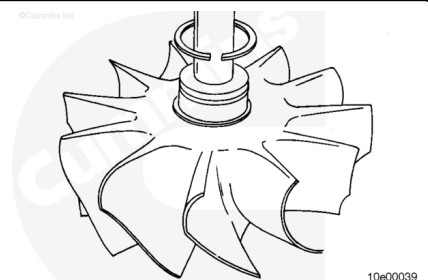
Turn the center housing over and repeat the snap ring and bearing installation for the compressor side.



Turn the center rotating housing over and install the thermal shield.

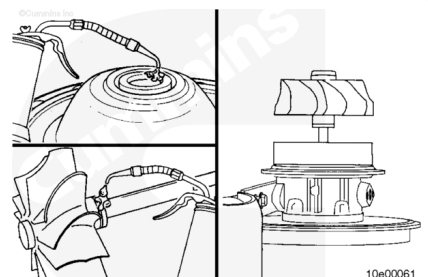


Install a new sealing ring in the groove in the turbine shaft.



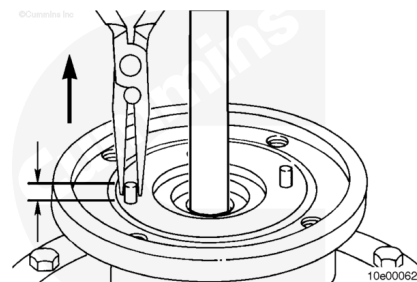
Lubricate the turbine shaft bore in the center housing with clean engine oil.

Position the seal ring so the split is opposite the oil flow. Lightly press on the turbine impeller while turning to aid in installation of the turbine shaft.



⚠ CAUTION ⚠

Use care when positioning the axial thrust plate pins. If crushing occurs, replace the pins.

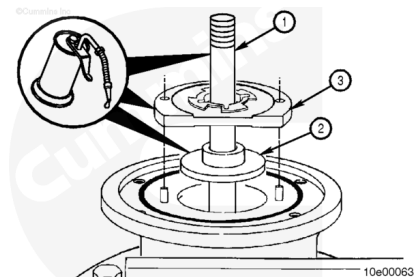


Set the turbine wheel in an appropriate support.

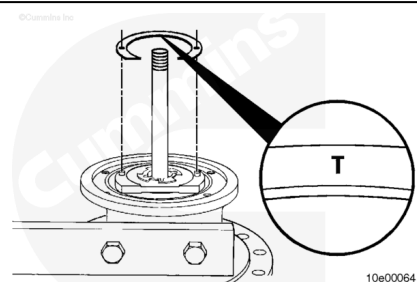
Use a pair of pliers to pull the axial thrust plate pins out to an approximate height of 4.83 mm [0.19 in].

Use clean oil to lubricate the turbine shaft (1), thrust ring (2), and the axial thrust plate (3).

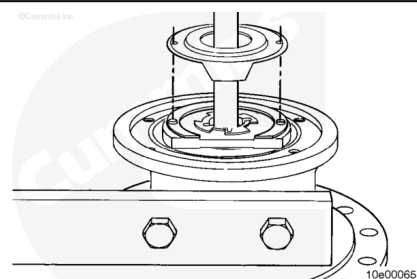
Install the thrust ring (2) and the axial thrust plate (3) onto the turbine shaft (1).



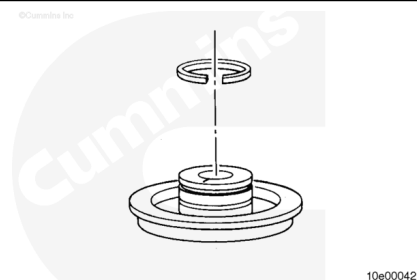
Install the adjustment shim with the letter T facing up.



Install the oil baffle.

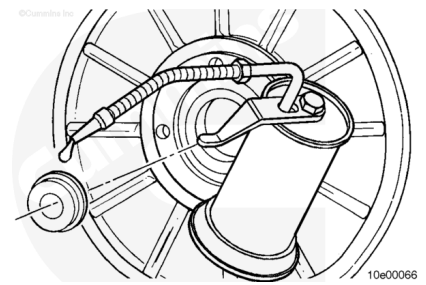


Install the sealing ring onto the diffuser plate nozzle vane.



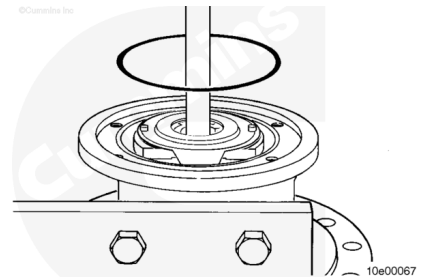
Use clean engine oil to lubricate the oil slinger seal ring.

Install the oil slinger into the diffuser plate.

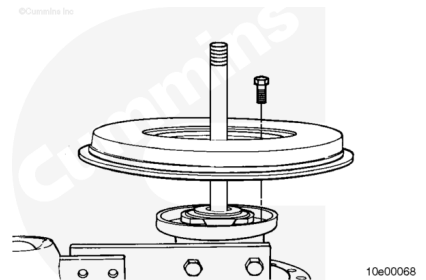


Use clean engine oil to lubricate the new o-ring.

Install the o-ring into the center housing rotating assembly compressor side.

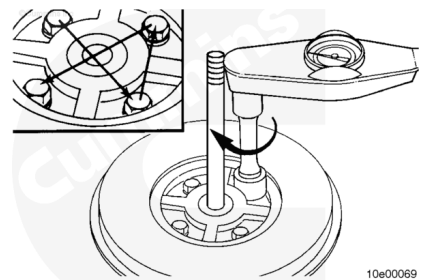


Place the diffuser plate on the center housing rotating assembly and install the capscrews.



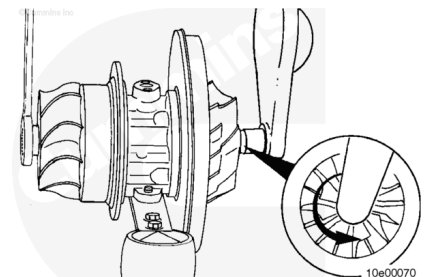
Tighten the capscrews in the sequence illustrated in the graphic.

Torque Value: 22 n•m [195 in-lb]



⚠ CAUTION ⚠

When using the original compressor impeller and shaft, the alignment marks must line up to maintain the balance. If the compressor impeller and turbine shaft are not aligned, vibration and premature turbocharger damage will occur.



Install and align the compressor impeller onto the turbine shaft.

Thread the locknut, flat side down, onto the turbine shaft in a **counterclockwise** direction.

Tighten the locknut.

Torque Value: 38 n•m [28 ft-lb]

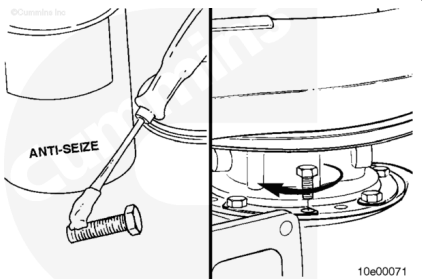
Place the center rotating housing assembly into the turbine housing.

Align the scribe marks on the turbine housing

Install the capscrews with new locking plates onto the turbine housing.

Tighten the capscrews.

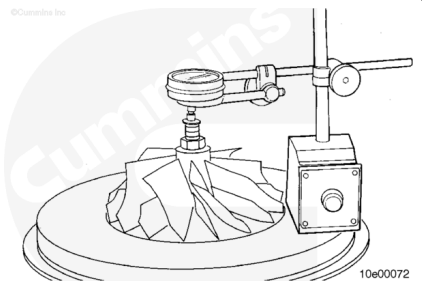
Torque Value: 22 n•m [195 in-lb]



Use a dial indicator to check the turbocharger end clearance.

Holset® HX82 Turbocharger End Clearance

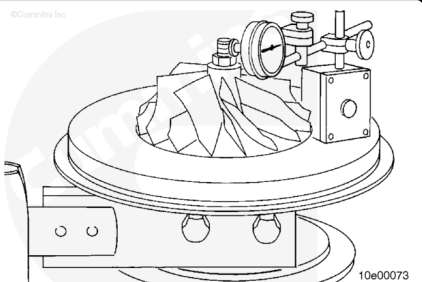
mm		in
0.025	MIN	0.001
0.152	MAX	0.006



Use a dial indicator to check the compressor impeller radial movement at the nose of the impeller.

Holset® HX82 Turbocharger Radial Movement

mm		in
0.254	MIN	0.010
0.787	MAX	0.031



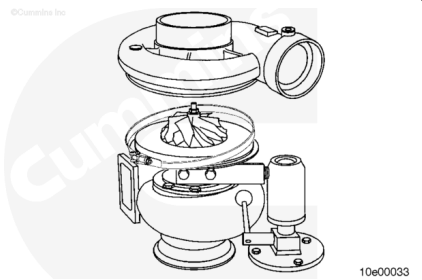
⚠ CAUTION ⚠

Use care when installing the compressor housing onto the turbocharger. The compressor impeller can be easily damaged.

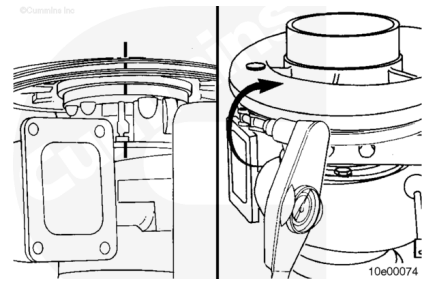
Carefully install the compressor housing on the center housing rotating assembly.

Align the scribe mark on the compressor housing with the mark on the center housing rotating assembly.

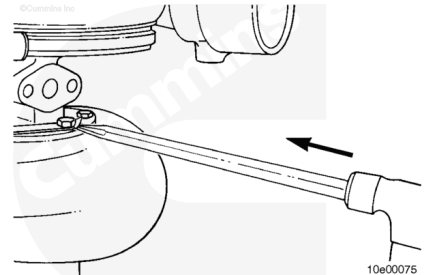
Install and tighten the V-band clamp.



Torque Value: 8.5 n•m [75 in-lb]



Use a hammer and chisel to bend the locking tabs over the cap screws on the turbine housing.



HX83 Turbocharger

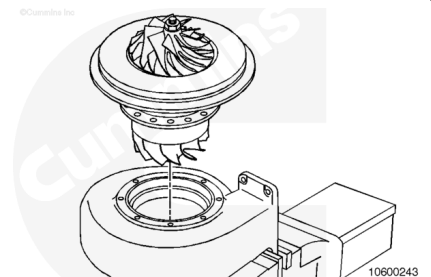
Install the center housing rotating assembly into the turbine housing.

Apply anti-seize compound to the threads of the new turbine housing cap screws.

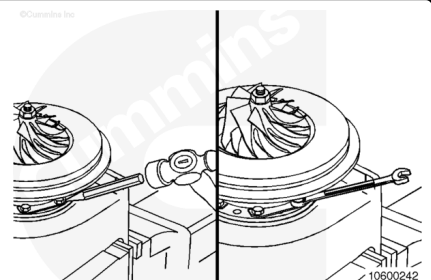
Install the new lockplates and turbine housing cap screws.

Tighten the cap screws.

Torque Value: 20 n•m [177 in-lb]



Use a hammer and chisel to bend the lockplate tabs over the turbine housing cap screws.

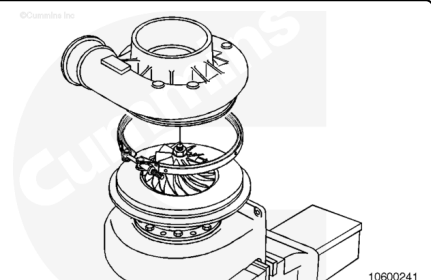


Loosely fit the V-band clamp over the center housing rotating assembly.

Install and align the compressor housing.

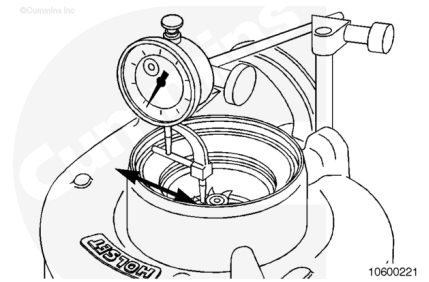
Tighten the V-band clamp.

Torque Value: 12 n•m [106 in-lb]



⚠ CAUTION ⚠

Insufficient compressor impeller-to-compressor housing clearance will cause serious engine damage.



Measure the radial clearance (side-to-side) at the compressor impeller nose using a dial gauge.

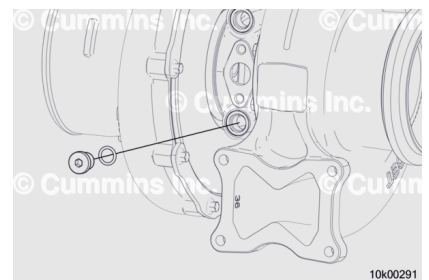
Compressor Impeller Radial Endplay

	mm		in
Holset® HX83 Turbocharger	0.254	MIN	0.01
	0.787	MAX	0.031

HE850 Turbocharger

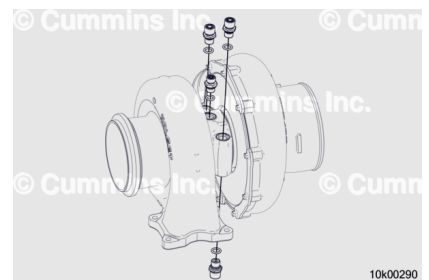
Install straight thread o-ring plug. Use new o-ring seal.

Torque Value: 34 n•m [25 ft-lb]



Install male unions. Use new o-ring seals.

Torque Value: 45 n•m [33 ft-lb]



Install reducing adapter (1). Use new o-ring seal.

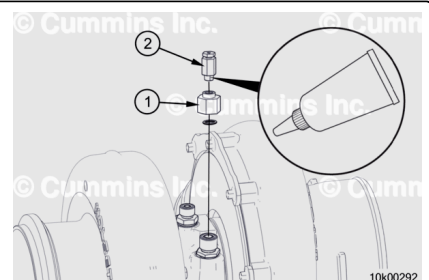
Torque Value: 45 n•m [33 ft-lb]

Apply Loctite® 577, Part Number 4919202, to coolant vent threads.

Install coolant vent (2).

Torque Value: 18 n•m [159 in-lb]

Tighten coolant vent screw (3).



Torque Value: 8 n•m [71 in-lb]

Last Modified: 17-Aug-2022
